

SQL Joins on Sales and Customer Tables

CODE:

```
CREATE DATABASE purchase;
```

```
USE purchase;
```

```
CREATE TABLE Sales (  
    SalesID INT AUTO_INCREMENT PRIMARY KEY,  
    ProductName VARCHAR(100) NOT NULL,  
    Qty INT NOT NULL,  
    DateOfPurchase DATE NOT NULL,  
    CustomerID INT NOT NULL  
);
```

```
CREATE TABLE Customer (  
    CustomerID INT PRIMARY KEY,  
    CustomerName VARCHAR(100) NOT NULL,  
    PhoneNumber VARCHAR(15)  
);
```

```
INSERT INTO Customer (CustomerID, CustomerName, PhoneNumber) VALUES  
(101, 'Ash', '123-456-7890'),  
(102, 'Boston', '987-654-3210'),  
(103, 'Clara', '456-789-0123');
```

```
INSERT INTO Sales (ProductName, Qty, DateOfPurchase, CustomerID) VALUES  
( 'Laptop', 10, '2023-01-05', 101),  
( 'Mouse', 50, '2023-01-15', 102),  
( 'Keyboard', 20, '2023-01-20', 101),  
( 'Laptop', 5, '2023-02-10', 103),  
( 'Mouse', 30, '2023-02-25', 101),  
( 'Keyboard', 15, '2023-03-03', 102);
```

```
SELECT  
    s.SalesID,  
    s.ProductName,  
    s.Qty,  
    s.DateOfPurchase,  
    c.CustomerName  
FROM Sales s  
INNER JOIN Customer c ON s.CustomerID = c.CustomerID;
```

```
SELECT  
    c.CustomerName,
```

```

        COALESCE(SUM(s.Qty), 0) AS TotalProductsPurchased
FROM Customer c
LEFT JOIN Sales s ON c.CustomerID = s.CustomerID
GROUP BY c.CustomerID, c.CustomerName;

```

```

SELECT
    s.ProductName,
    c.CustomerName,
    MIN(s.DateOfPurchase) AS FirstPurchaseDate
FROM Sales s
RIGHT JOIN Customer c ON s.CustomerID = c.CustomerID
GROUP BY s.ProductName, c.CustomerName;

```

output:

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	SalesID	ProductName	Qty	DateOfPurchase	CustomerName
▶	1	Laptop	10	2023-01-05	Ash
	2	Mouse	50	2023-01-15	Boston
	3	Keyboard	20	2023-01-20	Ash
	4	Laptop	5	2023-02-10	Clara
	5	Mouse	30	2023-02-25	Ash
	6	Keyboard	15	2023-03-03	Boston

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	ProductName	CustomerName	FirstPurchaseDate
▶	Mouse	Ash	2023-02-25
	Keyboard	Ash	2023-01-20
	Laptop	Ash	2023-01-05
	Keyboard	Boston	2023-03-03
	Mouse	Boston	2023-01-15
	Laptop	Clara	2023-02-10

<			
Result Grid			
Filter Rows:			
Export:			
Wrap Cell Content:			
	ProductName	CustomerName	FirstPurchaseDate
▶	Mouse	Ash	2023-02-25
	Keyboard	Ash	2023-01-20
	Laptop	Ash	2023-01-05
	Keyboard	Boston	2023-03-03
	Mouse	Boston	2023-01-15
	Laptop	Clara	2023-02-10